Exploratory Data Analysis

Introduction

The project will focus on analysing street crime data from four police forces, separated by four rural and four urban areas, over a four year period, ranging from January 2024 to January 2026. The dataset consists of monthly street crime records that were combined and cleaned during the preprocessing stage. The purpose of this exploratory data analysis (EDA) is to identify patterns in crime types, frequency and distribution across the selected police forces.

Visualisations will be used to highlight crime trends, regional differences, and urban versus rural patterns. The findings from this analysis will help identify two interesting police forces for further study in the next stage of the project.

Dataset Overview

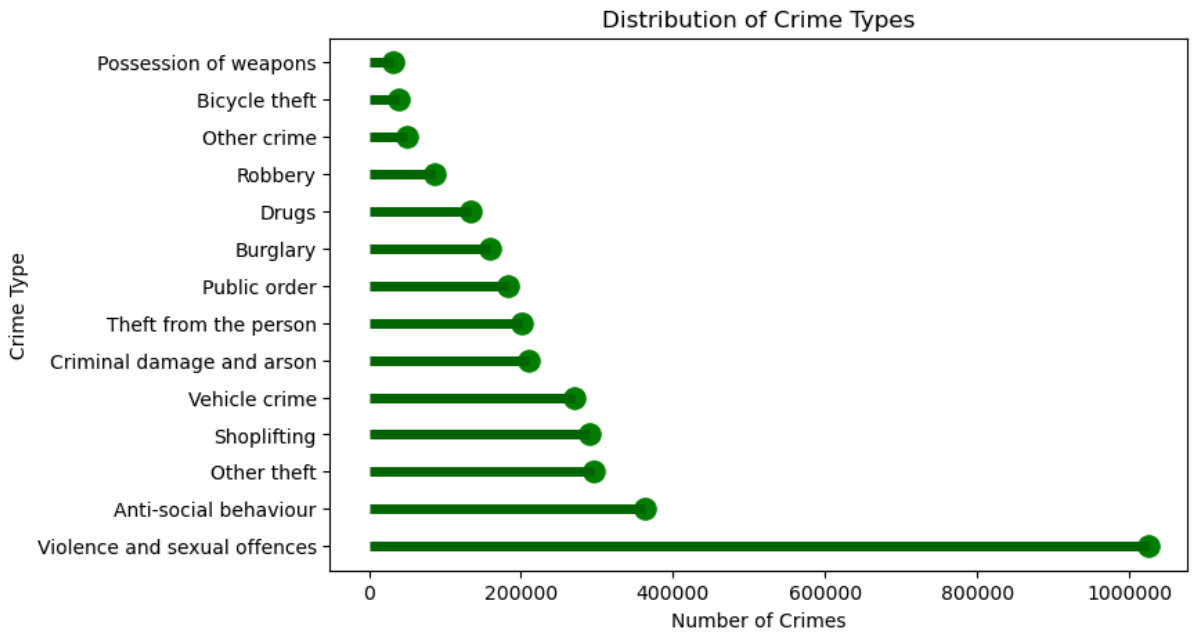
The dataset used for this analysis contains 3,337,712 rows and 13 columns representing recorded street crime incidents from four selected police forces between January 2024 and January 2026. The selected police forces are Devon & Cornwall Police, Metropolitan Police Service, North Yorkshire Police and West Midlands Police.

Each row represents an individual crime record, while some may not have a Crime ID, removing them could lead to an underestimation of crime levels, and the analysis focuses mainly on crime patterns rather than tracking individual cases. The columns contain information about the incident such as crime type, location coordinates, date, police force and crime outcome.

The dataset was prepared during the preprocessing stage where monthly files were combined, missing values were handled and data quality checks were performed to ensure consistency across the dataset.

Crime Distribution Overview

This section explores the overall distribution of crime types across the dataset. Understanding which crimes occur most frequently provides a general overview of crime patterns before comparing regions or analysing trends over time.

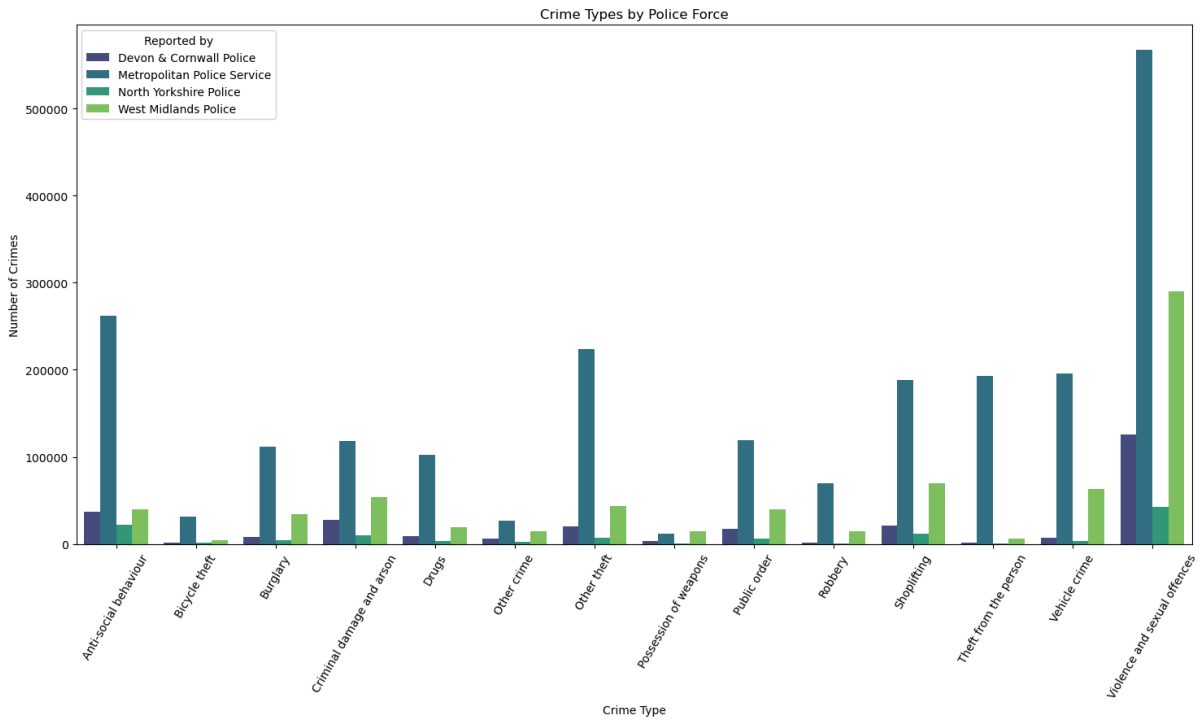


The results show that violence and sexual offences are the most frequently recorded crimes within the dataset, with 1,025,793 incidents, accounting for over 30% of total recorded crimes, dominating the crime category across the police forces. This is followed by anti-social behaviour, which represents a significant portion of reported incidents with 362,167 cases. Although these two are the most common crime types, there is a substantial gap between them, with violence occurring at nearly three times the frequency of anti-social behaviour. One possible explanation for this difference is that violent offences often include a wider range of incidents, such as assault or harassment, which may be reported more frequently and prioritised. In contrast, anti-social behaviour can go underreported or handled informally, which may result in lower recorded figures despite still being common in communities.

Several types of theft-related crimes appear prominently in the dataset, including other theft, shoplifting, vehicle crime, and theft from the person, indicating that property-related offences are common across the analysed regions which will be further analysed in the next sections.

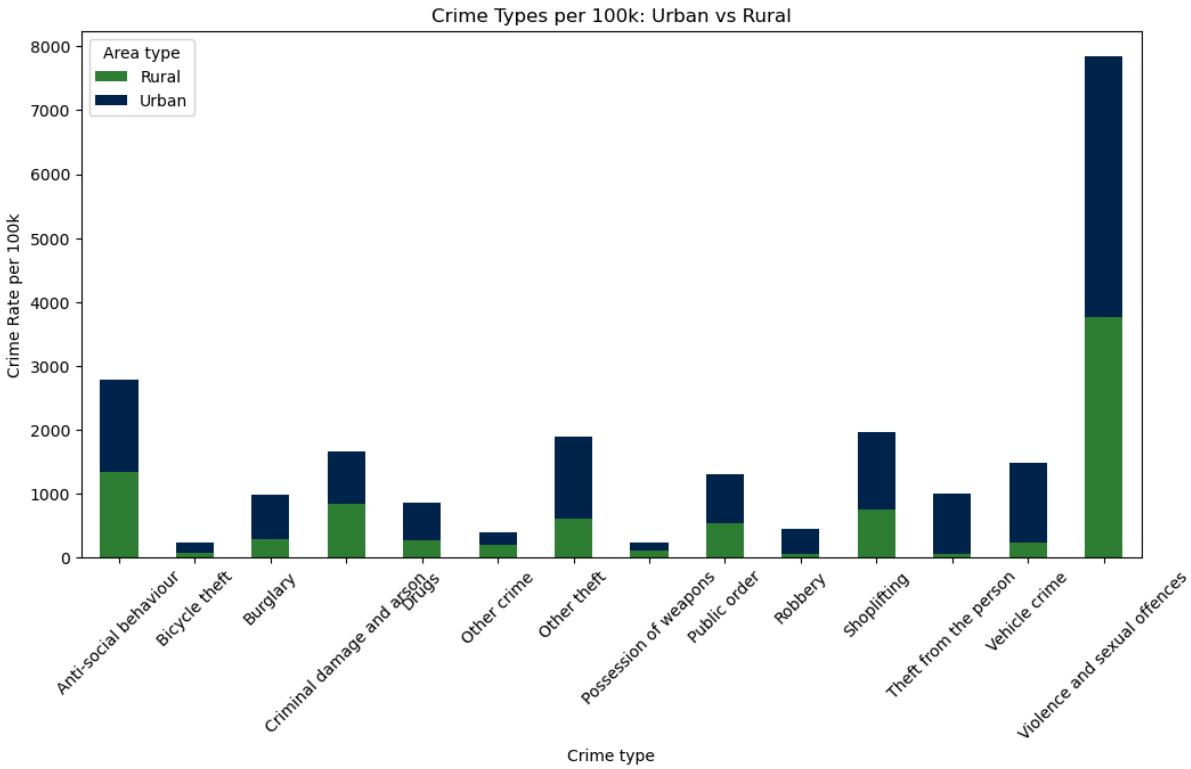
Crime Comparison Between Urban and Rural Police Forces

This section compares the total number of recorded crimes across the four selected police forces to identify regional differences in crime levels. By analysing the distribution of crimes between forces, it is possible to observe whether certain areas experience significantly higher or lower crime volumes.



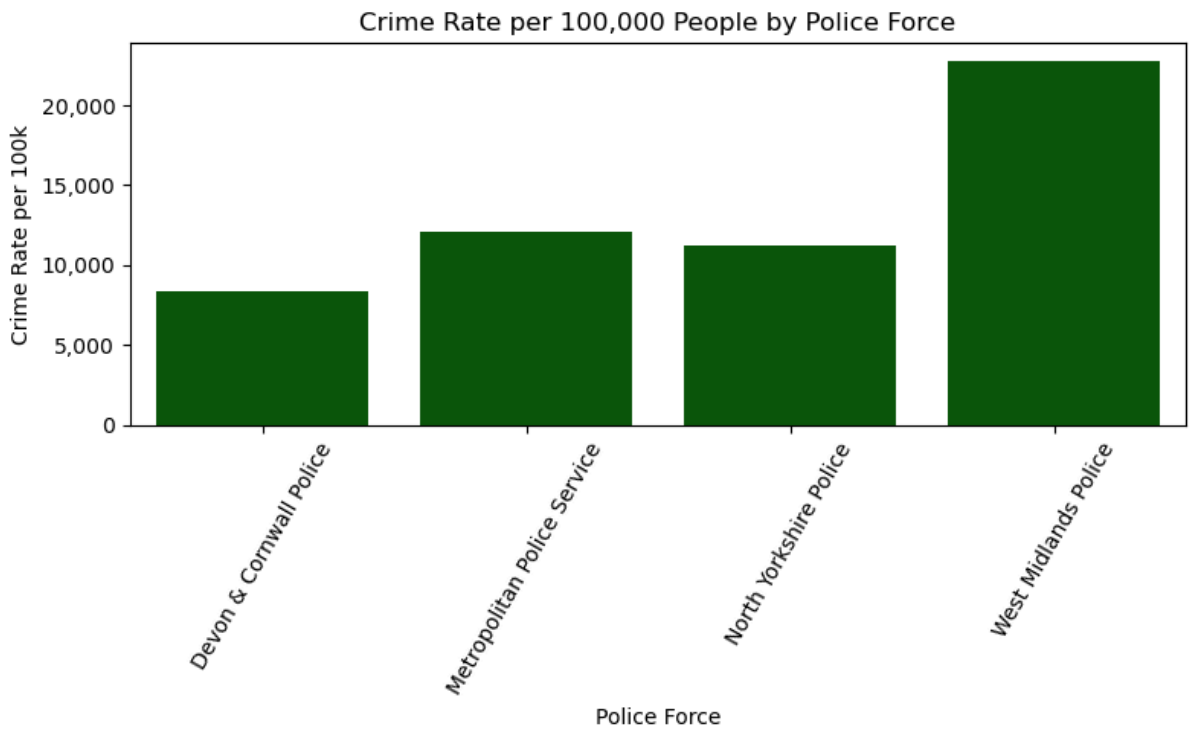
Initially, it can be seen that the Metropolitan Police Service area has crime totals reaching 2,223,054 crimes, with West Midlands Police coming in second with 708,277 total crimes. The record shows a noticeably higher number of incidents than others which may reflect differences in population size, urban density and economic activity within these areas, especially in the ‘Violence and sexual offences’ section. Visualising the distribution across regions enforces that ‘Violence and sexual offences’ remains the highest occurring crime in all cities, allowing stakeholders to make decisions on what to focus and invest more on as well as the region area.

Police forces located in more urbanised regions tend to report higher numbers of crimes overall, while forces covering more rural areas generally record lower totals.



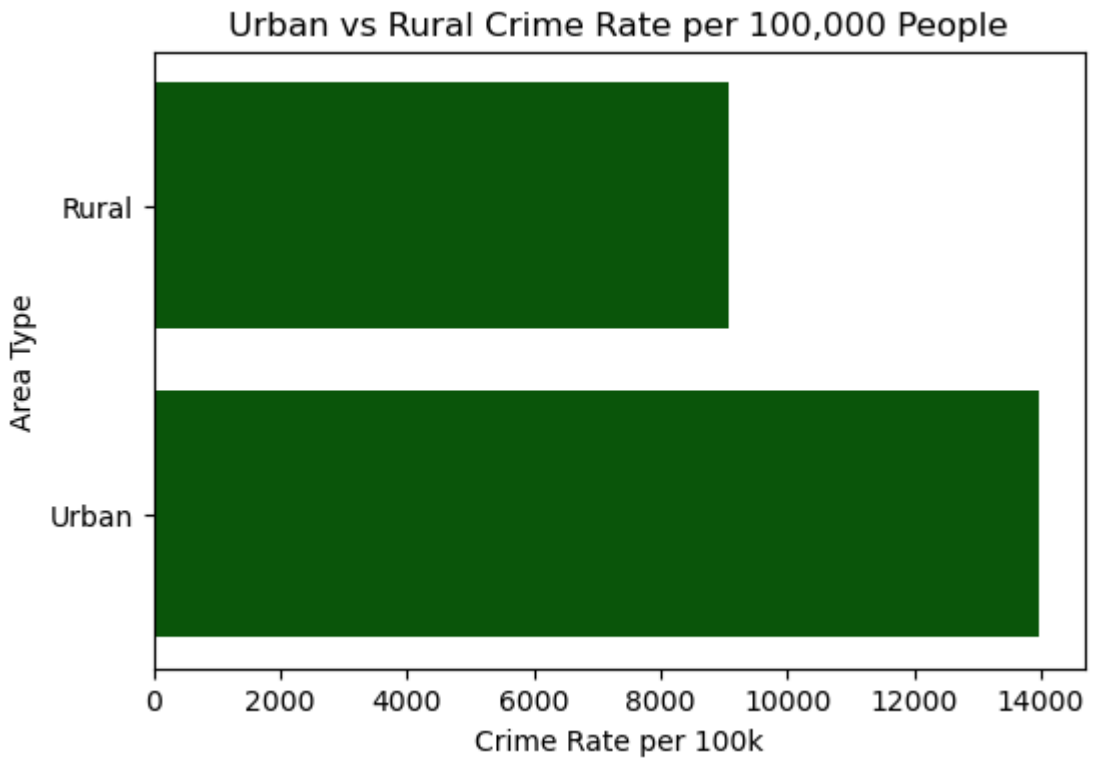
The comparison of crime types between urban and rural areas highlights clear differences in both the frequency and distribution of offences. Overall urban areas record higher crime rates across most crime categories, even after accounting for population differences. Violence and sexual offences remain the most prevalent crime type in both environments, though the rate is slightly higher in urban areas, indicating a greater concentration of violent incidents in more densely populated regions.

However, ‘Criminal damage and arson’ appears slightly higher in rural areas compared to urban regions, which may be due to factors such as vandalism in less supervised areas, damage to rural property, or incidents involving farmland and isolated buildings where there is less surveillance. Overall, while urban areas experience a wider concentration of crime, rural areas may still see relatively higher levels of certain property-related offences.



After doing further analysis, when adjusting for population, although the West Midlands Police comes third in ranking for total crime, the rate of crime to population comes in first with 22,751 crimes per 100,000 people, implying that, despite having a much smaller population than the Metropolitan Police Service, it has the highest concentration of recorded crime.

North Yorkshire Police, which represents a more rural region, recorded the lowest total number of crimes of 118,265 and a comparatively lower crime rate of around 11,179 per 100,000 people. Devon Cornwall Police also recorded relatively fewer crimes overall, indicating lower crime levels compared to the more urbanised police force areas.

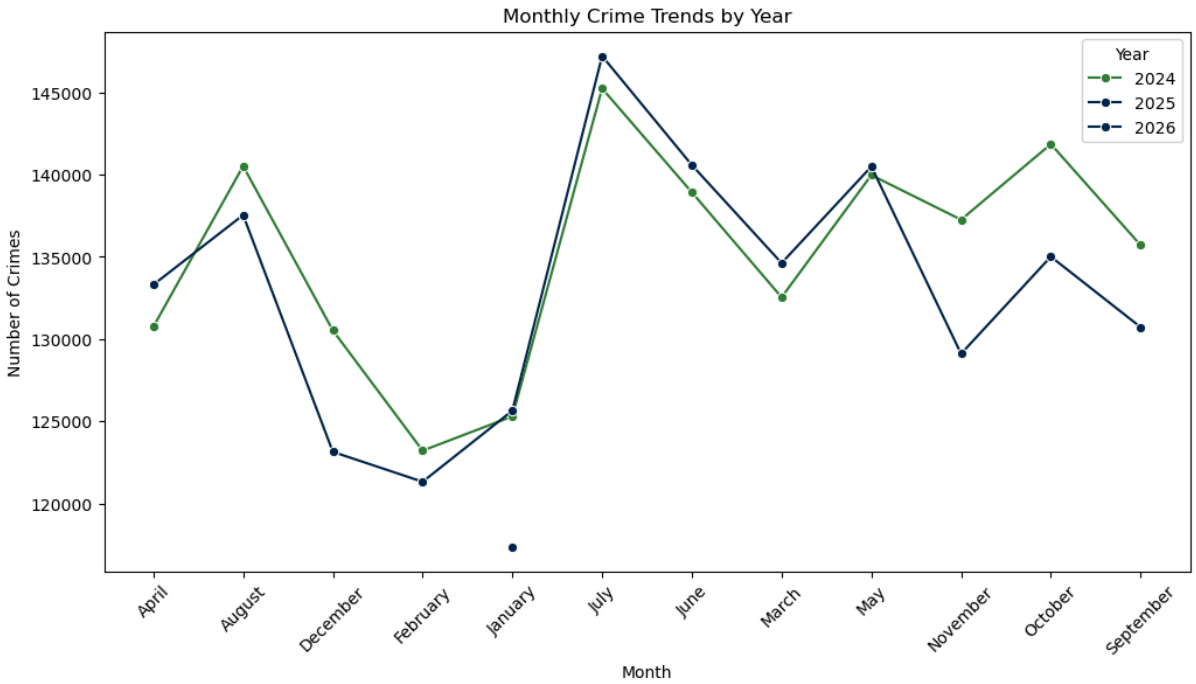


Urban areas still record a noticeable higher crime rate compared to rural regions continuing to show a greater concentration of crime relative to its population size. This pattern may be influenced by several factors commonly associated with urban environments, including higher population density, greater economic activity and more opportunities for certain types of crime such as theft or violent offences. In contrast, rural regions generally have lower population density, which may reduce opportunities for crime to occur. As a result, even when population differences are accounted for, urban areas may still experience proportionally higher levels of recorded crime.

This analysis provided valuable insights for stakeholders when comparing crime patterns between urban and rural areas. By adjusting crime totals relative to population size, stakeholders are able to gain a more accurate understanding of crime risk rather than relying solely on raw crime counts. This allows policymakers and local authorities agencies to better identify regions where crime is proportionally higher and may require additional resources. Overall, this analysis supports more informed decision-making when developing strategies tailored to the differing crime patterns found in urban and rural environments.

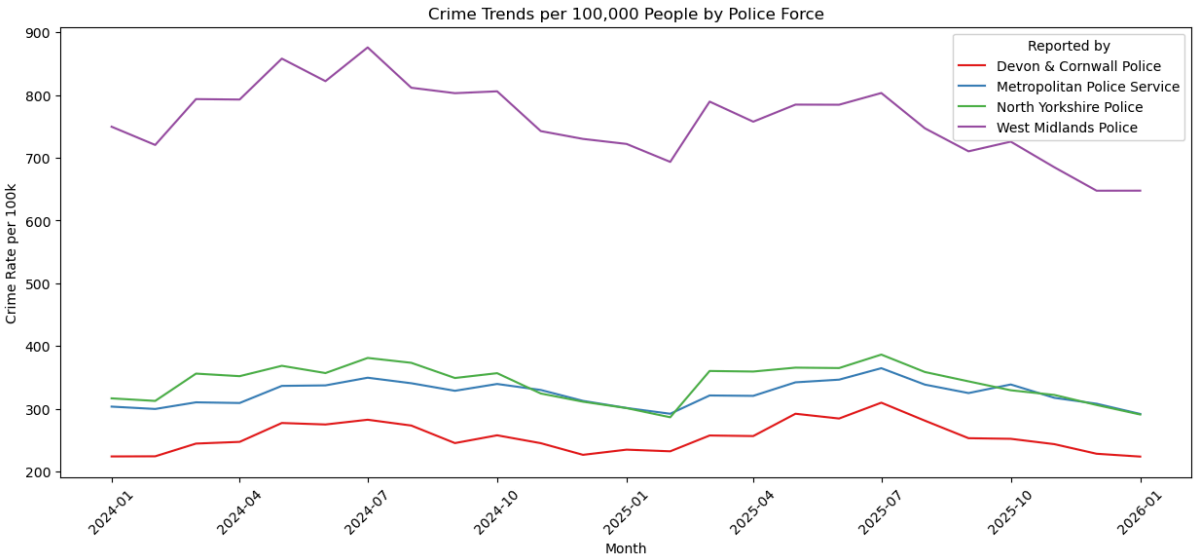
Crime trends over time

Analysing trends over time allows better insights on whether crime rates are affected by the area or if there are seasonal changes that take place during the different quarters of the year.



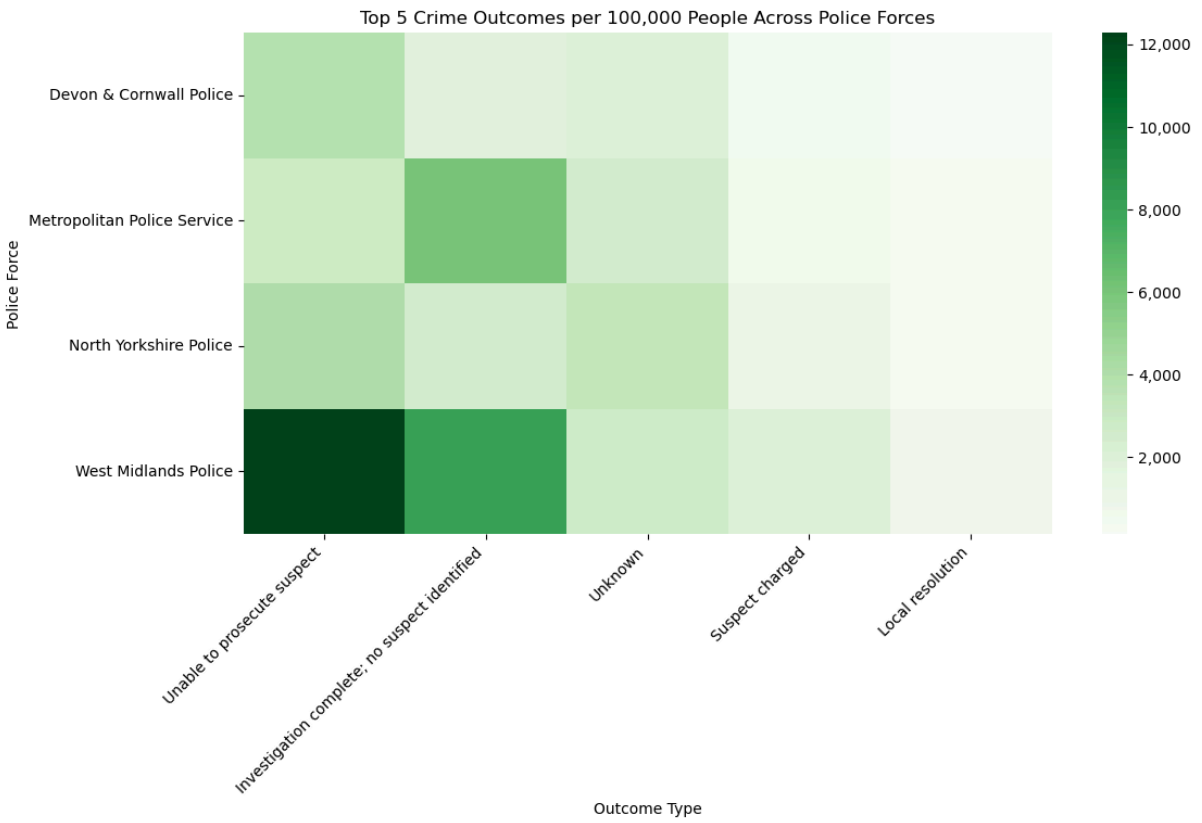
The monthly crime total shows noticeable seasonal variation across the analysed period. Similarly, 2024 and 2025 generally show an increase in crime levels during the middle of the year, with both highest being recorded in July, at 145,228 and 147,198 respectively, while lowest levels occur during the winter months, particularly January and February, sitting at 125,295 and 121,319 respectively. This indicates a possible seasonal influence on criminal activity with the incline during summer months and decline in the winter months. The data for 2026 currently only includes January with low activity at 117,376, reflecting the likeness of incomplete data for that year.

Overall, identifying these monthly patterns can help stakeholders have a better understanding and detect trends to when crime levels tend to increase and may support more effective resource allocation and preventative strategies during periods of higher criminal activity.



The yearly comparison of crime across the four police forces show clear differences between regions, with some fluctuation in crime levels over time. The West Midlands Police consistently shows records of the highest levels of crime, reflecting the characteristics of a densely populated urban area. The two peaks shown in all four areas are also found to occur during summer which further enforces seasonal relative crimes. While crime totals vary slightly between years across all forces, the overall pattern remains consistent and an insignificant sign of increase over the course of the years, suggesting seasonality may have more of an impact rather than the year.

Crime Outcome Analysis



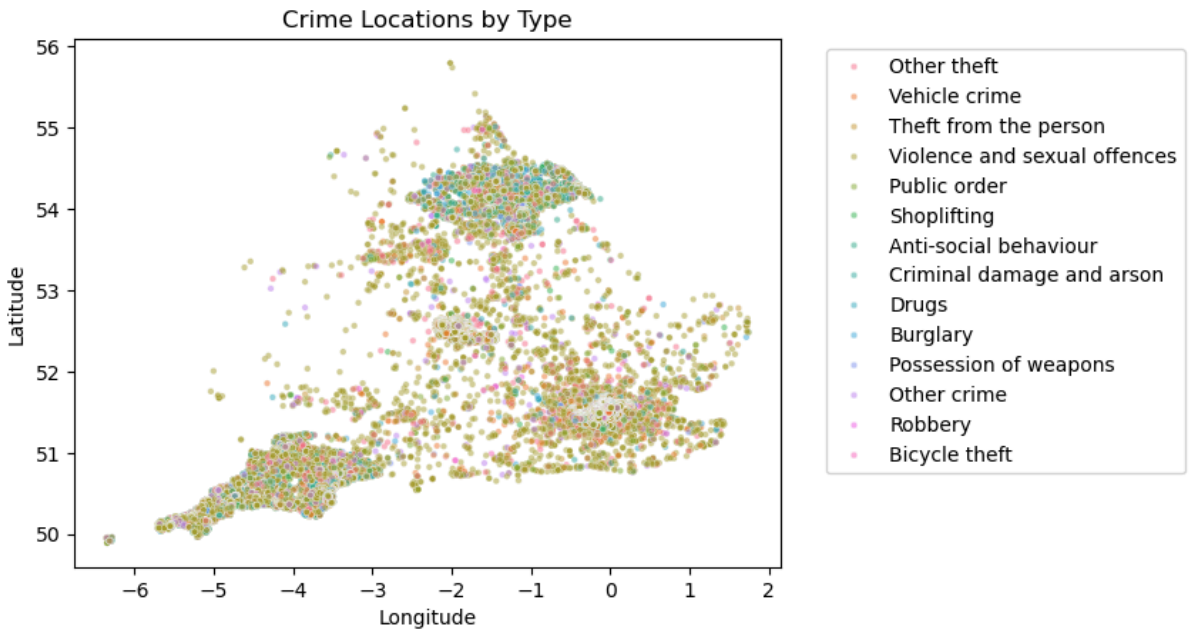
The heatmap illustrates the top five crime outcomes per 100,00 people across the four police forces, allowing for a fair comparison between areas with very different population sizes. Overall, the most common outcome is ‘unable to prosecute suspects’ indicating that a large proportion of recorded crimes do not result in a suspect being successfully identified. The Westland Midlands Police show the highest rate reflecting the challenges of investigating crimes in densely populated urban environments where crime volumes are higher.

The outcome category ‘Unknown’ also appears across all police forces where the category typically indicates cases where the final investigation outcome was not recorded or was

unavailable at the time the data was compiled. Higher levels reflect incomplete case updates and ongoing investigations making it more difficult to fully assess how effectively certain incidents are resolved. Recognising the presence of this category highlights the importance of improving data recording and case tracking processes to ensure policymakers to better evaluate investigation and monitor crime resolution rates.

When comparing urban and rural areas, a similar pattern can be observed. The urban forces generally show higher outcome rates per 100,00 people, particularly for outcomes related to unsuccessful investigations, suggesting the complexity of crime in urban settings, where higher population density and increased mobility can make investigations more difficult. In contrast, the rural police forces display lower rates, although the same dominant outcomes still appear. This suggests that while crime volumes are lower in rural regions, the overall pattern of investigation outcomes remains similar, highlighting the broader challenges in resolving certain types of crime across both urban and rural policing environments.

Crime Mapping



The map illustrates the geographic distribution of recorded crime incidents across England, showing how crimes are spread based on their latitude and longitude coordinates. A clear concentration of incidents can be observed in the major urban areas, particularly around London and the West Midlands. These regions display a dense clustering of points, indicating a higher volume of recorded crimes. This pattern likely reflects higher population density and increased economic activity.

In contrast, rural areas such as North Yorkshire and parts of Devon and Cornwall show fewer and more widely dispersed crime incidents. The points in these regions are more spread out, indicating lower overall crime volumes and greater geographic distance between individual incidents. While a variety of crime types are represented across the map, they appear throughout the country rather than being confined to specific regions. Overall, the map

highlights the strong relationship between urbanisation and crime concentration, with densely populated areas experiencing higher levels of recorded crime compared to rural regions

Key Insights

The analysis shows that ‘Violence and sexual offences’ represent the largest proportion of recorded crimes despite separating into years and regional areas, the crime type was the top incident across all categories. ‘Criminal damage and arson’ was the only category within the rural category that surpassed the urban category indicating property-related offences may be more prevalent in rural areas, potentially due to factors such as less surveillance or isolated properties, therefore suggesting potential areas of concern that stakeholders may wish to focus on when developing targeted crime prevention in rural areas. While every other category was more dominant in the urban areas, this is expected due to the urban density of big cities and the dense population where police forces struggle to resolve or prevent these problems, however still can be beneficial in knowing which areas to be focused on the most.

When comparing crime levels between police forces, the Metropolitan Police Service records the highest number of crimes overall which reflects the size and density of London. However, when adjusting for population size, West Midlands Police records the highest crime rate per 100,000 people, indicating that crime risk can appear different when population differences are taken into account. This suggests that the West Midlands area may experience a proportionally higher level of crime relative to its population compared to the other regions. Between the rural areas, Devon & Cornwall has a higher level of crime compared to North Yorkshire Police, suggesting that crime distribution can still vary significantly between rural regions depending on factors such as distribution of population and economic conditions.

Analysis of crime trends over time shows a clear seasonal pattern where levels tend to rise and increase during summer months as displayed in 2024 as well as 2025, particularly in July and August. All years had a decline during winter months during January and February. This consistent seasonal trend may be influenced by seasonal social activity and outdoor interactions during the warmer months, hence reflecting the need for effective policing preparations during peak periods.

The outcome for the majority of the cases came to conclude with no suspect found or prosecuted, further implying that stakeholders could focus on improving investigation resources and preventative policing strategies to reduce the occurrence of such crimes.

Chosen Police Forces

After conducting analysis between rural and urban areas, West Midlands Police and North Yorkshire Police were selected for further analysis due to their great distribution and representation of two contrasting policing environments within the dataset. The West Midlands Police covers a highly urbanised and densely populated area and records the

highest number of crimes despite having a smaller population compared to The Metropolitan Police Service. In contrast, North Yorkshire Police serves a largely rural region with a smaller population. By selecting these two police forces, the analysis can more clearly compare how crime patterns differ between urban and rural areas without having too great of a difference in population. This allows for a better understanding of how factors such as population density, environment and geographic distribution influence crime levels and types.